

India Carbon Market Model

Technical Documentation

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1 Model Overview

This model simulates India’s Carbon Credit Trading Scheme (CCTS), the country’s emerging carbon pricing mechanism based on a **baseline-and-credit market** framework

1.1 Economic Framework

Key structural features include:

- **Small open economy:** Product prices are fixed by international markets.
- **Firm optimization:** Firms minimize total costs by adopting abatement technologies and adjust output in response to changes in unit production costs.
- **Market clearing:** The carbon price adjusts to equate aggregate allowance demand and supply.
- **Subsector heterogeneity:** Cost structures and output flexibility vary across sub-sectors.

2 Core Model Mechanics

2.1 Firm Decision-Making

Facing a carbon price P_C , each firm makes two interrelated decisions.

2.1.1 Technology Adoption

A firm adopts all abatement technologies with marginal abatement cost below the carbon price:

$$\text{Adopt technology } t \text{ if } MAC_t \leq P_C \quad (1)$$

Total emission intensity reduction is:

$$A_i(P_C) = \sum_{t \in J_i(P_C)} \gamma_{t,i} \quad (2)$$

Negative-cost abatement options are excluded from the simulation to avoid treating them as fully and frictionlessly available, given that their adoption may be constrained by technological maturity, financing, or organizational barriers in practice.

2.1.2 Output Adjustment

Production responds to proportional unit cost changes:

$$Q_i^* = Q_{i,0} \times (1 - \delta_i)^{\varepsilon_{ss}} \quad (3)$$

where

$$\delta_i = \frac{\Delta u_i}{u_{\text{baseline},i}}$$

2.2 Net Carbon Position and Market Equilibrium

Each firm's net carbon position is:

$$N_i(P_C) = Q_i^*(P_C) \times [r_{\text{int},i} - A_i(P_C)] \quad (4)$$

The equilibrium carbon price satisfies:

$$\sum_i N_i(P_C^*) = 0 \quad (5)$$

3 Scenario Design and Summary

The model is evaluated under multiple scenarios reflecting uncertainty in baseline emissions and technology adoption.

3.1 Core Scenarios

- **Scenario A: Autonomous Technical Progress** Baseline emission intensity is reduced exogenously by 2%.
- **Scenario B: Baseline** No autonomous efficiency improvement; negative-cost options are excluded.
- **Scenario C: Slow Technology Adoption** Effective abatement potential is reduced to 30% of its technical maximum.

3.2 Price-Based Policy Scenarios

- **Price floor:** Minimum prices of \$5 and \$10 per tCO₂ applied to Scenario A.
- **Price ceiling:** Maximum prices of \$15 and \$20 per tCO₂ applied to Scenario C.

4 Price Control Mechanisms

4.1 Price Floor

If the unconstrained equilibrium price falls below the minimum level:

$$P_C = P^{\text{floor}} \quad (6)$$

Excess allowance supply is implicitly cancelled.

4.2 Price Ceiling

If the unconstrained equilibrium price exceeds the ceiling:

$$P_C = P^{\text{ceiling}} \quad (7)$$

Residual imbalance reflects unmet abatement at the capped price.

5 Subsector Parameters

5.1 Baseline Prices and Supply Elasticities

Sector	Subsector	Baseline price (u_{baseline} USD/ton)	Supply elasticity (ε)
ALUMINIUM	REFINERY	450	0.70
ALUMINIUM	SECONDARY ALUMINIUM	2000	1.20
ALUMINIUM	SMELTER	2400	0.50
CEMENT	CLINKERIZATION	95	0.50
CEMENT	COMPOSITE	90	0.70
CEMENT	GRINDING	80	0.80
CEMENT	ORDINARY PORTLAND CEMENT	95	0.60
CEMENT	PORTLAND POZZOLANA CEMENT	90	0.65
CEMENT	WHITE CEMENT	180	0.50
CHLOR-ALKALI	CHLOR-ALKALI	260	0.80
IRON & STEEL	IRON & STEEL	850	0.60
PAPER & PULP	AGRO BASED PLANT	580	1.00
PAPER & PULP	INTEGRATED PLANT	650	0.70
PAPER & PULP	RCF BASED PLANT	540	0.90
PAPER & PULP	SPECIALITY PAPER PLANT	760	1.10
PETROCHEMICAL	PETROCHEMICAL	1000	0.80
REFINERY PETROLEUM	REFINERY PETROLEUM	750	0.70
TEXTILE	COMPOSITE	1700	1.20
TEXTILE	FIBRE	2000	1.00
TEXTILE	PROCESSING	1800	1.40
TEXTILE	SPINNING	1900	1.30

Table 1: Subsector baseline prices and supply elasticities

Hidden assumptions: Baseline product prices are treated as exogenous and equal to long-run average unit production costs, implying zero economic profits and the absence of firm-level markups, consistent with a small open economy framework. Consequently, output adjustments are driven solely by supply-side cost changes; incorporating firm-level markups, more precise price–cost relationships, or empirically estimated elasticities would allow for a richer representation of firm behavior and can be accommodated in future extensions. Demand-side responses are not explicitly modeled.

6 Model Inputs and Outputs

6.1 Model Inputs

The model is calibrated using detailed firm- and subsector-level inputs that characterize production, emissions, technology options, and behavioral responses. Key inputs include:

- **Firm-level production and emissions:** Baseline output ($Q_{i,0}$), baseline emission intensity ($e_{\text{baseline},i}$), and target emission intensity ($e_{\text{target},i}$) for each covered firm.
- **Abatement technology data:** Facility-level marginal abatement cost curves (MACCs), including technology-specific marginal abatement costs (MAC_t) and associated emission reduction potentials ($\gamma_{t,i}$).
- **Baseline prices and unit costs:** Subsector-specific baseline unit production costs (u_{baseline}), treated as exogenous product prices under the small open economy assumption.
- **Supply elasticities:** Subsector-level supply elasticities (ε) governing firms' output responses to proportional unit cost changes.
- **Policy parameters:** Emission intensity targets, scenario-specific technology effectiveness parameters, and price-based policy constraints (price floors and price ceilings).

6.2 Model Outputs

The model produces outcomes at the market, sectoral, and firm levels, including equilibrium carbon prices, emission reductions, production responses, and the distribution of buyers and sellers in the carbon market.